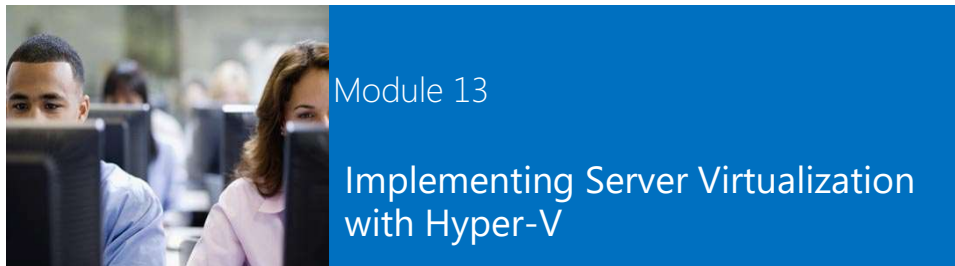


Microsoft® Official Course



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Windows Azure

Windows Azure is an open and flexible cloud platform that enables you to quickly build, deploy and manage applications across a global network of Microsoft-managed datacenters.

Build applications using any language, tool or framework. You can integrate your public cloud applications with your existing IT environment.

GLOBAL
With 8 data centers worldwide, and a worldwide Content Delivery Network, you can build applications that provide the best experience even to the most remote users.

ANY LANGUAGE
Windows Azure allows you to use any Windows Framework you're used to, or build applications using .NET, PHP, Java, Ruby, Python and Ruby. Client libraries are available for C#, Java, and JavaScript.

OPEN PROTOCOLS
Windows Azure features and services are exposed using open REST protocols.

CONNECTED
Use the Windows Azure cloud computing capabilities to deliver hybrid solutions that run across the cloud and on-premise. Expand your data center into the cloud with Virtual Networking.

SELF-HEALING
Windows Azure provides automatic OS and service patching, built-in network fault tolerance and self-healing for hardware failure. Windows Azure delivers 99.99% monthly SLA.

SELF-SERVICE
It's a fully automated self-service platform that allows you to provision resources on-demand.

ELASTIC RESOURCES
Quickly scale your resources based on your needs. Pay only for the resources your application uses.

REVENUE READY
Backed by industry certifications for security and compliance, ISO 27001, LAM, HIPAA BAA and EU Model Clauses.

CLOUD SERVICES

VIRTUAL MACHINES

WEB SITES

MOBILE SERVICES

MEDIA SERVICES

Cloud Services consist of Internet-facing data and Web services that can be deployed on-demand. Web sites can either be deployed to the Windows Azure cloud or to your own on-premise infrastructure to enable peak of Web sites. All roles can access data stored in other services.

Mobile Services is a hybrid cloud service running on Windows Azure servers that offers developers a simple, reliable, scalable, integrative service and pricing architecture for mobile applications. You can run code on the server or mobile devices (e.g., when doing CRUD operations).

Media Services provides a collection of services for encoding and processing streaming media such as video and audio, and your high-resolution content can be stored in the multiple hot data for playback on a variety of devices. Upload to HLS, DASH and protocols.

COMPUTE	DATA SERVICES	APP SERVICES	NETWORKING	STORE
Virtual Machines Virtual Machines (VMs) are the core of Windows Azure. They are used to run applications and services. VMs can be created from a template or from a custom image. They can be scaled up or down as needed.	Table Storage Table Storage is a NoSQL database service that stores data in a flat, tabular format. It is used for storing large amounts of unstructured data.	Web Sites Web Sites is a service that allows you to build and host web applications. It supports a wide range of programming languages and frameworks.	Network Interface Cards (NICs) Network Interface Cards (NICs) are used to connect VMs to the network. They provide a way to manage network traffic and security.	SQL Database SQL Database is a fully managed relational database service. It provides a simple way to create, manage, and scale a SQL database.
Web Role Web Roles are used to run web applications. They are created from a template or from a custom image. They can be scaled up or down as needed.	Blob Storage Blob Storage is a service for storing large amounts of unstructured data, such as text or binary files. It is used for storing data that is not structured in a table.	Mobile Services Mobile Services is a hybrid cloud service that allows you to build and host mobile applications. It supports a wide range of programming languages and frameworks.	Virtual Network Virtual Network is a service that allows you to create and manage a virtual network. It provides a way to manage network traffic and security.	SQL Azure SQL Azure is a fully managed relational database service. It provides a simple way to create, manage, and scale a SQL database.
Worker Role Worker Roles are used to run background tasks. They are created from a template or from a custom image. They can be scaled up or down as needed.	Queue Storage Queue Storage is a service for storing messages. It is used for storing data that is not structured in a table.	Web Services Web Services are used to build and host web applications. They support a wide range of programming languages and frameworks.	Network Security Network Security is a service that allows you to create and manage a virtual network. It provides a way to manage network traffic and security.	SQL Data Warehouse SQL Data Warehouse is a fully managed relational database service. It provides a simple way to create, manage, and scale a SQL database.

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Types de virtualisation

Desktop virtualization inclut les technologies suivantes :

- Hyper-V (local)
 - VDI (Virtual Desktop Infrastructure)
-
- Différence entre Desktop Virtualization et Presentation Virtualization :
 - Desktop Virtualization : une machine virtuelle (OS) par utilisateur
 - Presentation Virtualization : une machine virtuelle (OS) pour plusieurs utilisateurs

Virtual Machine Hardware

Matériel simulé par défaut :

- BIOS
- Memory
- Processor
- IDE Controller 0 and 1
- SCSI Controller
- Synthetic Network Adapter
- COM 1 and 2
- Diskette Drive

Matériel pouvant être ajouté :

- SCSI Controller (up to 4)
- Network Adapter
- Legacy Network Adapter
- Fibre Channel adapter
- RemoteFX 3D video adapter

Generation 2 Virtual Machines

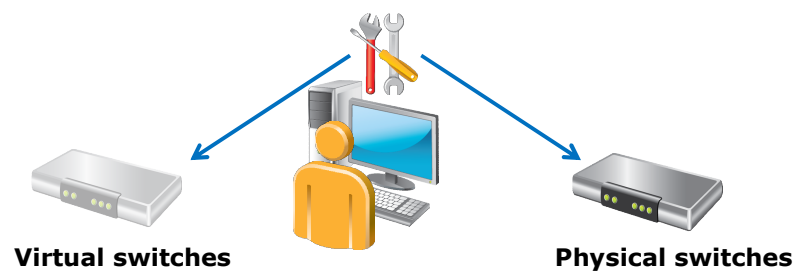
- Génération 1 : Fournit le même matériel virtuel à la machine virtuelle que dans les versions précédentes d'Hyper-V
- Génération 2 : Apporte de nouvelles fonctionnalités
 - PXE boot
 - Boot à partir d'un disque dur virtuel SCSI
 - Boot à partir d'un lecteur DVD virtuel SCSI
 - Secure Boot (démarrage signé avec une clé chiffrée)
 - Support du UEFI firmware

Virtual Machine Integration Services

- Les services d'intégration comprennent:
 - Arrêt du système d'exploitation
 - Synchronisation de l'heure
 - L'échange de données
 - Sauvegarde (instantanés)
- Le mode de session amélioré permet une connexion à une machine virtuelle similaire à celle du bureau à distance car elle:
 - Permet la redirection de périphérique
 - Utilise un presse-papiers partagé
 - Permet la redirection de dossiers

Virtual Switch Extensions

- Permet aux fournisseurs tiers de créer des switches virtuels pouvant être ajoutés à Hyper-V.
- Ils se configurent comme les switches physiques et apportent les fonctionnalités suivantes : surveillance, filtrage et transfert de paquets



NIC Teaming?

- Renforce la redondance et libère de la bande passante
- Est pris en charge par les hôtes, ainsi que par les machines virtuelles
- NIC Teaming et les machines virtuelles :
 - Nécessite plusieurs cartes réseaux virtuelles
 - Doit être activé sur un adaptateur réseau virtuel
 - Peut-être implémenté dans le système d'exploitation de la machine virtuelle (si pris en charge)

Lab: Implementing Server Virtualization with Hyper-V

- Exercise 1: Installing the Hyper-V Role onto a Server
- Exercise 2: Configuring Virtual Networking
- Exercise 3: Creating and Configuring a Virtual Machine
- Exercise 4: Using Virtual Machine Checkpoints

Logon Information

Virtual machine	20410D-LON-HOST1
User name	Administrator
Password	Pa\$\$w0rd

Estimated Time: 70 minutes

Lab Scenario

Your assignment is to configure the infrastructure service for a new branch office.

To use the server hardware that is available currently at branch offices more effectively, your manager has decided that all branch office servers will run as virtual machines. You must now configure a virtual network and a new virtual machine for these branch offices.

Lab Review

- What type of virtual network switch would you create if you want to allow the virtual machine to communicate with the LAN that is connected to the Hyper-V virtualization server?
- How can you ensure that no single virtual machine uses all of the available bandwidth that the Hyper-V virtualization server provides?
- What Dynamic Memory configuration task was not possible on previous versions of Hyper-V, but which you can now perform on a virtual machine that is hosted on the Hyper-V role on a Windows Server 2012 server?

Module Review and Takeaways

- Review Questions
- Best Practices
- Common Issues and Troubleshooting Tips
- Tools

Course Evaluation

